

Week 1 Homework and Weekly Quiz

Signals • LTI Systems • Convolution • Eigenfunctions • DTFT

Instructor version with complete solutions

Audience	ECE seniors + first-year graduate students
Suggested homework effort	90–120 minutes core + 20–30 minutes graduate stretch
Suggested quiz time	15 minutes
Text alignment	Oppenheim & Schaffer, Discrete-Time Signal Processing — LTI systems and DTFT topics

Teaching intent. The core problems emphasize mathematical reasoning and indexing discipline. The graduate-stretch items ask for proof, convergence, or generalization beyond a single numerical example.

Part I — Homework Problems

Show enough work that another engineer can follow your indexing, assumptions, and reasoning. Use Python only where a problem explicitly asks for it or as a check after the analytic work.

1. Impulse and step representations [8 points]

The sequence $x[n]$ is $x[-1]=3$, $x[0]=-2$, $x[2]=4$, and zero elsewhere.

- Express $x[n]$ as a sum of scaled, shifted unit samples.
- Let $v[n]=u[n+1]-u[n-2]$. List the nonzero samples of $v[n]$ and sketch it.
- Express $\delta[n+2]$ using two shifted unit steps.

2. System-property classification [16 points]

For each system below, determine whether it is linear, time invariant, causal, and BIBO stable. Give a one- or two-sentence justification for every “no.”

- $y[n]=2x[n-2]$
- $y[n]=x^2[n]$
- $y[n]=n x[n]$
- $y[n]=x[n]+0.5x[n-1]$

3. Finite convolution and support [10 points]

Let $x[n]=\{1, -1, 2\}$ for $n=0, 1, 2$ and $h[n]=\{2, 1\}$ for $n=0, 1$.

- Predict the index range over which $y[n]=x[n]*h[n]$ can be nonzero.
- Compute the full convolution by hand.
- Verify the result with `np.convolve`.

4. Impulse response, causality, stability, and step response [12 points]

An LTI system has $h[n]=(0.75)^n u[n]$.

- Is the system causal? Explain from $h[n]$.
- Is the system BIBO stable? Show the relevant sum.
- Find the response to $x[n]=u[n]$. Give a closed-form expression for $n \geq 0$.
- What is the limiting output value as $n \rightarrow \infty$?

5. Sinusoid through an LTI system [12 points]

An LTI system has $H(e^{j\omega})=1+0.5e^{-j\omega}$. The input is $x[n]=2\cos(\pi n/3+\pi/4)$.

- Evaluate $H(e^{j\pi/3})$ in rectangular form.
- Find its magnitude and phase.
- Write the steady-state output $y[n]$ in cosine form.

6. DTFT shifting and a spectral zero [10 points]

Let $x[n]=(0.5)^n u[n]$ and $y[n]=x[n-2]-x[n]$.

a) Find $X(e^{j\omega})$.

b) Use the time-shift property to find $Y(e^{j\omega})$.

c) Evaluate $Y(e^{j0})$ and explain what that value says about the sum of $y[n]$.

7. Difference equation to frequency response [12 points]

A causal system satisfies $y[n] - 0.8y[n-1] = x[n] - x[n-1]$, with zero initial conditions.

a) Derive $H(e^{j\omega})$.

b) Evaluate H at $\omega=0$ and $\omega=\pi$.

c) Based only on those values and the structure of the numerator, is the system more “low-pass-like” or “high-pass-like”? Explain.

d) Use `scipy.signal.freqz` to check your reasoning.

8. Direct DTFT evaluation [10 points]

Let $x[n] = \{1, 2, -1\}$ for $n=0, 1, 2$.

a) Write $X(e^{j\omega})$ explicitly.

b) Evaluate $X(e^{j\omega})$ at $\omega=0$, $\pi/2$, and π .

c) Use the 2π -periodicity property to state $X(e^{j5\pi/2})$ without recomputing the sum.

9. BIBO stability proof for an LTI system [10 points] — Graduate stretch

Assume $|x[n]| \leq B_x$ for all n and that $\sum_k |h[k]| < \infty$. Starting from $y[n] = \sum_k x[k]h[n-k]$, prove that $y[n]$ is bounded for all n . State an explicit bound B_y .

Then explain in one sentence why $h[n] = (0.99)^n u[n]$ is stable even though its impulse response lasts forever.

Part II — Homework Solutions

1. Impulse and step representations

- a) $x[n] = 3\delta[n+1] - 2\delta[n] + 4\delta[n-2]$. The sign inside $\delta[n-n_0]$ is tied to the location n_0 .
- b) $u[n+1]$ turns on at $n=-1$ and $u[n-2]$ turns on at $n=2$. Therefore $v[n]=1$ for $n=-1, 0, 1$ and 0 otherwise.
- c) $\delta[n+2] = u[n+2] - u[n+1]$.

2. System-property classification

- A. $y[n] = 2x[n-2]$: linear, time invariant, causal, BIBO stable. It is a fixed gain and fixed delay.
- B. $y[n] = x^2[n]$: nonlinear, time invariant, causal, BIBO stable. Squaring violates superposition, but $|x| \leq B$ implies $|y| \leq B^2$.
- C. $y[n] = n x[n]$: linear, time varying, causal, not BIBO stable. A bounded input $x[n]=1$ produces $y[n]=n$, which is unbounded.
- D. $y[n] = x[n] + 0.5x[n-1]$: linear, time invariant, causal, BIBO stable. Its impulse response is $\delta[n] + 0.5\delta[n-1]$, whose absolute sum is 1.5.

3. Finite convolution and support

- a) x is supported on $0 \dots 2$ and h on $0 \dots 1$, so y is supported on $0 \dots 3$. The full-convolution length is $3+2-1=4$.
- b) $y[0] = 1 \cdot 2 = 2$. $y[1] = 1 \cdot 1 + (-1) \cdot 2 = -1$. $y[2] = (-1) \cdot 1 + 2 \cdot 2 = 3$. $y[3] = 2 \cdot 1 = 2$. Therefore $y = \{2, -1, 3, 2\}$.
- c) `np.convolve([1,-1,2],[2,1])` returns $[2, -1, 3, 2]$.

4. Impulse response, causality, stability, and step response

- a) Causal: $h[n]=0$ for $n<0$ because of $u[n]$.
- b) Stable: $\sum_n |h[n]| = \sum_{n=0}^{\infty} (0.75)^n = 1/(1-0.75) = 4 < \infty$.
- c) For $n \geq 0$, $y[n] = \sum_{k=0}^n (0.75)^k = (1 - 0.75^{n+1}) / (1 - 0.75) = 4(1 - 0.75^{n+1})$. For $n < 0$, $y[n] = 0$.
- d) $\lim_{n \rightarrow \infty} y[n] = 4$.

5. Sinusoid through an LTI system

At $\omega_0 = \pi/3$, $e^{-j\pi/3} = 1/2 - j\sqrt{3}/2$. Therefore $H = 1 + 0.5(1/2 - j\sqrt{3}/2) = 1.25 - j0.4330$.

$|H| = \sqrt{(1.25^2 + 0.4330^2)} = \sqrt{1.75} \approx 1.3229$. $\angle H \approx -0.3335$ rad (-19.1°).

For a real sinusoid through a real LTI system, $y[n] = 2|H|\cos(\pi n/3 + \pi/4 + \angle H)$. Thus $y[n] \approx 2.6458 \cos(\pi n/3 + 0.4519)$.

6. DTFT shifting and a spectral zero

- a) $X(e^{j\omega}) = 1/(1 - 0.5e^{-j\omega})$ because $|0.5| < 1$.
- b) $x[n-2] \leftrightarrow e^{-j2\omega}X(e^{j\omega})$, so $Y(e^{j\omega}) = (e^{-j2\omega} - 1)/(1 - 0.5e^{-j\omega})$.
- c) $Y(e^{j0}) = (1 - 1)/(1 - 0.5) = 0$. Since $Y(e^{j0}) = \sum_n y[n]$ when the sum converges, the samples of $y[n]$ sum to zero.

7. Difference equation to frequency response

Taking the DTFT (or substituting a complex exponential): $Y - 0.8e^{-j\omega}Y = X - e^{-j\omega}X$.

Therefore $H(e^{j\omega}) = (1 - e^{-j\omega})/(1 - 0.8e^{-j\omega})$.

At $\omega=0$, $H=0$. At $\omega=\pi$, $H=(1-(-1))/(1-0.8(-1))=2/1.8\approx 1.111$.

The zero at DC and nonzero response at π make the system high-pass-like. A freqz plot should confirm stronger high-frequency response.

8. Direct DTFT evaluation

a) $X(e^{j\omega})=1+2e^{-j\omega}-e^{-j2\omega}$.

b) $X(1)=X(e^{j0})=1+2-1=2$. At $\omega=\pi/2$: $e^{-j\pi/2}=-j$ and $e^{-j\pi}=-1$, so $X=1-2j+1=2-2j$. At $\omega=\pi$: $X=1-2-1=-2$.

c) $5\pi/2=\pi/2+2\pi$, so $X(e^{j5\pi/2})=X(e^{j\pi/2})=2-2j$.

9. BIBO stability proof for an LTI system

$$|y[n]| = |\sum_k x[k]h[n-k]| \leq \sum_k |x[k]| |h[n-k]| \leq B_x \sum_k |h[n-k]|.$$

A shift does not change the absolute sum, so $\sum_k |h[n-k]| = \sum_m |h[m]|$. Therefore $|y[n]| \leq B_x \sum_m |h[m]| = B_y < \infty$ for every n .

For $h[n] = (0.99)^n u[n]$, the absolute sum is $1/(1-0.99)=100$, which is finite. Infinite duration does not imply instability; lack of absolute summability does.

Part III — Weekly Quiz (Student Version)

Suggested time: 15 minutes. Total: 20 points. Calculator not required.

1. [2 pts] Which sequence equals $\delta[n-3]$?

- A. $u[n-3]-u[n-4]$ B. $u[n+3]-u[n+2]$ C. $u[n-2]-u[n-3]$ D. $u[n]-u[n-3]$

Answer: _____

2. [2 pts] The system $y[n]=x[n-2]+x[n+2]$ is:

- A. linear, time invariant, causal B. linear, time invariant, noncausal C. nonlinear, time invariant D. linear, time varying

Answer: _____

3. [2 pts] A length-5 sequence convolved with a length-4 sequence has what full-convolution length?

Show the number and one sentence of reasoning.

Answer: _____

4. [3 pts] Let $x[n]=\{1,2\}$ and $h[n]=\{1,-1\}$, both beginning at $n=0$. Compute $y[n]=x*h$.

Answer: _____

5. [2 pts] Why are complex exponentials called eigenfunctions of LTI systems?

Answer in one or two sentences.

Answer: _____

6. [2 pts] If $X(e^{j\omega})$ is a DTFT, what is $X(e^{j(\omega+2\pi)})$?

Explain why in one line.

Answer: _____

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7. [4 pts] For $y[n]=0.5y[n-1]+x[n]$, derive $H(e^{j\omega})$. Then evaluate H at $\omega=0$ and $\omega=\pi$.

Answer: _____

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8. [3 pts] Based on Question 7, does the system emphasize low or high normalized frequencies? Explain without plotting.

Answer: _____

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Part IV — Weekly Quiz Solutions

1. Unit-sample identity

A. $\delta[n-3]=u[n-3]-u[n-4]$.

2. System properties

B. The rule is linear and a fixed combination of shifts, so it is time invariant. It is noncausal because $y[n]$ depends on the future value $x[n+2]$.

3. Convolution length

$5+4-1=8$ samples. Full linear convolution adds the support lengths minus one.

4. Short convolution

$[1,2]*[1,-1]=[1,1,-2]$. Specifically $y[0]=1$, $y[1]=-1+2=1$, $y[2]=-2$.

5. Eigenfunction meaning

For input $e^{j\omega n}$, every LTI system produces $H(e^{j\omega})e^{j\omega n}$. The function's shape/frequency is unchanged; only a complex scale factor is applied.

6. DTFT periodicity

$X(e^{j(\omega+2\pi)})=X(e^{j\omega})$ because $e^{-j(\omega+2\pi)n}=e^{-j\omega n}e^{-j2\pi n}=e^{-j\omega n}$ for integer n .

7. First-order frequency response

Rearrange: $y[n]-0.5y[n-1]=x[n]$. Thus $H(e^{j\omega})=1/(1-0.5e^{-j\omega})$. At $\omega=0$, $H=2$. At $\omega=\pi$, $H=1/(1+0.5)=2/3$.

8. Qualitative interpretation

Low frequencies are emphasized because the response magnitude is larger at $\omega=0$ than at $\omega=\pi$. The denominator becomes smallest when $e^{-j\omega}$ is near $+1$.

Instructor scoring notes

For Questions 5, 6, and 8, award most credit for correct reasoning even if notation is imperfect. On Question 7, separate algebra/setup credit from the two endpoint evaluations. The quiz intentionally checks conceptual understanding more than arithmetic speed.